Master's Thesis Proposal

A Systematic Quality Assurance Framework for Evaluating the Usability and Reliability of the ODME Tool for Aviation Scenario Modeling

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1 Introduction

The Operation Domain Modeling Environment (ODME) is a specialized software tool used for constructing and analyzing system models within the aviation domain. Such tools play a critical role in supporting system design, analysis, and decision-making processes, where correctness and usability directly influence the reliability of derived models.

In safety-critical domains such as aviation, even minor inconsistencies in modeling tools can propagate into incorrect system representations, potentially leading to flawed analysis and decisions. Therefore, it is not sufficient to ensure only implementation correctness; it is equally important to validate whether the tool behaves reliably from a user perspective and supports intuitive, consistent, and error-free interaction [3].

The initial phase of this work focused on code-level unit testing to explore the internal structure and assess the testability of the ODME system. This investigation provided insights into architectural characteristics such as tight coupling and reliance on static components, which limit the effectiveness of purely code-level testing approaches in capturing real-world usage issues [1].

Based on these findings and further academic discussion, the research direction has been refined toward the development of a **systematic Quality Assurance (QA) framework**. This framework emphasizes evaluating the usability and reliability of the ODME tool from an end-user and application-oriented perspective, ensuring that it effectively supports real-world modeling tasks in aviation contexts.

2 Problem Statement

Although the ODME tool provides a wide range of functionalities for model construction and analysis, there is currently no structured framework to evaluate its usability and reliability in practical aviation modeling scenarios.

Specifically, the following challenges exist:

- Lack of systematic validation of feature correctness under real user workflows
- Limited assessment of usability for complex modeling tasks
- Absence of structured evaluation of model generation reliability
- Uncertainty regarding alignment between tool functionality and domain-specific requirements

In aviation-related applications, these limitations introduce risks such as incorrect model construction, inefficient workflows, and reduced confidence in system outputs. Therefore, a structured QA framework tailored to ODME is necessary to ensure its reliability and effectiveness [2].

3 Research Objectives

The primary objectives of this thesis are:

- To design a systematic Quality Assurance framework for evaluating the ODME tool
- To derive structured test cases based on functional specifications

- To evaluate the usability of the ODME interface in supporting modeling workflows
- To assess the correctness and reliability of model generation processes
- To identify defects, inconsistencies, and usability challenges
- To provide recommendations for improving the reliability and usability of the tool

4 Research Questions

This study aims to address the following research questions:

1. How effectively does ODME support real-world modeling workflows in aviation applications?
2. What usability challenges arise during interaction with the ODME interface?
3. How reliable is the tool in generating correct models based on user inputs?
4. To what extent does user interface design influence modeling accuracy and efficiency?
5. How can a structured QA framework improve confidence in ODME outputs?

5 Scope of Work

5.1 Feature-Based Validation

Validation of ODME functionalities based on updated feature documentation.

5.2 UI and Interaction Testing

Evaluation of user interactions including menus, toolbars, and workflows.

5.3 Use Case Evaluation

Execution and validation of a real-world aviation model generation scenario.

5.4 Usability Assessment

Evaluation of usability aspects such as efficiency, learnability, and error prevention.

6 Methodology

The research will follow a structured approach:

1. Analyze updated ODME feature documentation
2. Derive structured test scenarios and test cases
3. Perform systematic UI-based testing

4. Execute real-world use case scenarios
5. Document observed behavior and deviations
6. Analyze usability issues and interaction patterns
7. Evaluate the correctness and reliability of generated models
8. Formulate conclusions and recommendations

7 Expected Contributions

This thesis is expected to provide:

- A systematic QA framework tailored for aviation modeling tools
- A structured evaluation of usability and reliability of ODME
- Identification of design and interaction issues affecting modeling accuracy
- Practical recommendations for improving tool usability and robustness
- Validation of the tool through a real-world aviation use case

8 Previous Work

The initial phase of this work involved implementing unit tests across multiple components of the ODME system to understand its architecture and testability. This analysis revealed structural limitations such as tight coupling and reliance on static dependencies, which restrict the effectiveness of code-level testing for UI-driven systems.

These findings motivated the transition toward a higher-level QA framework focusing on usability and reliability evaluation.

9 Expected Outcomes

- A comprehensive set of QA test cases
- Documented usability evaluation results
- Identification of defects and inconsistencies
- Validation of a real-world modeling use case
- Recommendations for improving the ODME tool

10 Conclusion

This thesis aims to establish a systematic Quality Assurance framework for evaluating the ODME tool from a usability and reliability perspective. By focusing on real-world aviation modeling scenarios, the study seeks to bridge the gap between system functionality and practical usability, ultimately contributing to improved confidence in model-driven decision-making processes.

References

[1] Ammann, P., & Offutt, J. (2016). *Introduction to Software Testing*. Cambridge University Press.

[2] ISO/IEC 25010:2011. *Systems and software engineering – Systems and software Quality Requirements and Evaluation (SQuaRE).*

[3] Leveson, N. (2011). *Engineering a Safer World: Systems Thinking Applied to Safety.* MIT Press.